

**JAB WE MET**
A Covid inoculation drive for the 15-18 age group in Daryaganj, Delhi, Jan. 8

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**Call it a booster or precaution dose, but with the launch of its latest vaccination programme on January 10, India continues its battle against Covid-19. And it couldn't have started any sooner, for the highly contagious Omicron variant is driving up the number of cases in the pandemic's third wave. Close to 57.5 million people are eligible for the third shot, among them 27.5 million people above the age of 60, 10 million healthcare workers and 20 million frontline personnel. Eligibility can be checked on CoWIN, and a prior consultation with doctors is advisable.**

That done, all one needs is a mobile number or an identity card to take the third shot. A booster dose is meant to elicit a robust antibody response. However, while the Government of India-approved 'prevention' doses are vital for the vulnerable, virologists and other medical researchers also believe it is crucial to assess the cross-immunity generated by mixing available vaccines. They recommend that safety profiles be well established before administering different vaccines as prevention doses.

Apart from Covishield and Covaxin, seven other vaccines, including an mRNA vaccine that is easier to produce and quick to launch, have been approved. The mandatory two doses are expected to be given to all in the country in the next 4-5 months. The vaccines seem remarkably effective against all variants in preventing hospitalisation and saving lives. But long-term data on their use is not available yet. The vaccines have been approved on 'reasonable basis' on information available in little over a year with Emergency Use Authorisation (EUA).

"None of the vaccines, including the mRNA vaccines, are protecting against infections," says Dr Rakesh Mishra, director at the Tata Institute for Genetics and Society, Bengaluru. "The advantage with the third wave is that we have very large seropositivity, of about 80 per cent or more, and many of the vulnerable enjoy a fairly high degree of immune protection with one or two doses, though they may not be protected against infection."

Though research data is scanty, there is enough to show that antibodies decline six months after taking two shots of a vaccine. Hence the need for a booster dose. However, as Mishra points out, "Boosters have limited value, which is why they are called precaution doses. We do not have data to tell us if those taking it enjoy a specific percentage of protection. It provides a marginal top-up of antibodies, but we cannot say anything with certainty yet. To call it boosters, we should have data on how exactly it works. We do not have it now. What we know is the vaccines are safe and it is good to provide extra protection to the relatively more vulnerable."

## COCKTAIL CONUNDRUM

Though the Centre has for the moment ruled against mixing vaccines, a mixed dose of Covishield and Covaxin can certainly boost the spike protein-neutralising antibodies and enhance vaccine effectiveness even against Omicron, says Dr D. Nageshwar Reddy, founder-chairman of the Asian Institute of Gastroenterology (AIG) Hospitals and its adjunct Asian Healthcare Foundation. A study done by AIG apparently confirms this, as none of the participants showed any adverse effect. "When the first and second doses were from different vaccines, the spike protein antibody response was four times higher than that after two doses of the same vaccine," he says. Spike protein-neutralising antibodies kill the virus and reduce overall infectivity.

The AIG study also revealed that 87 per cent of the participants—who were not vaccinated and had never tested positive—had Covid antibodies. "This means our population might have developed significant antibodies against Covid because of the huge Delta wave we endured," says Reddy.

Epidemiologists, however, are sceptical about implementing